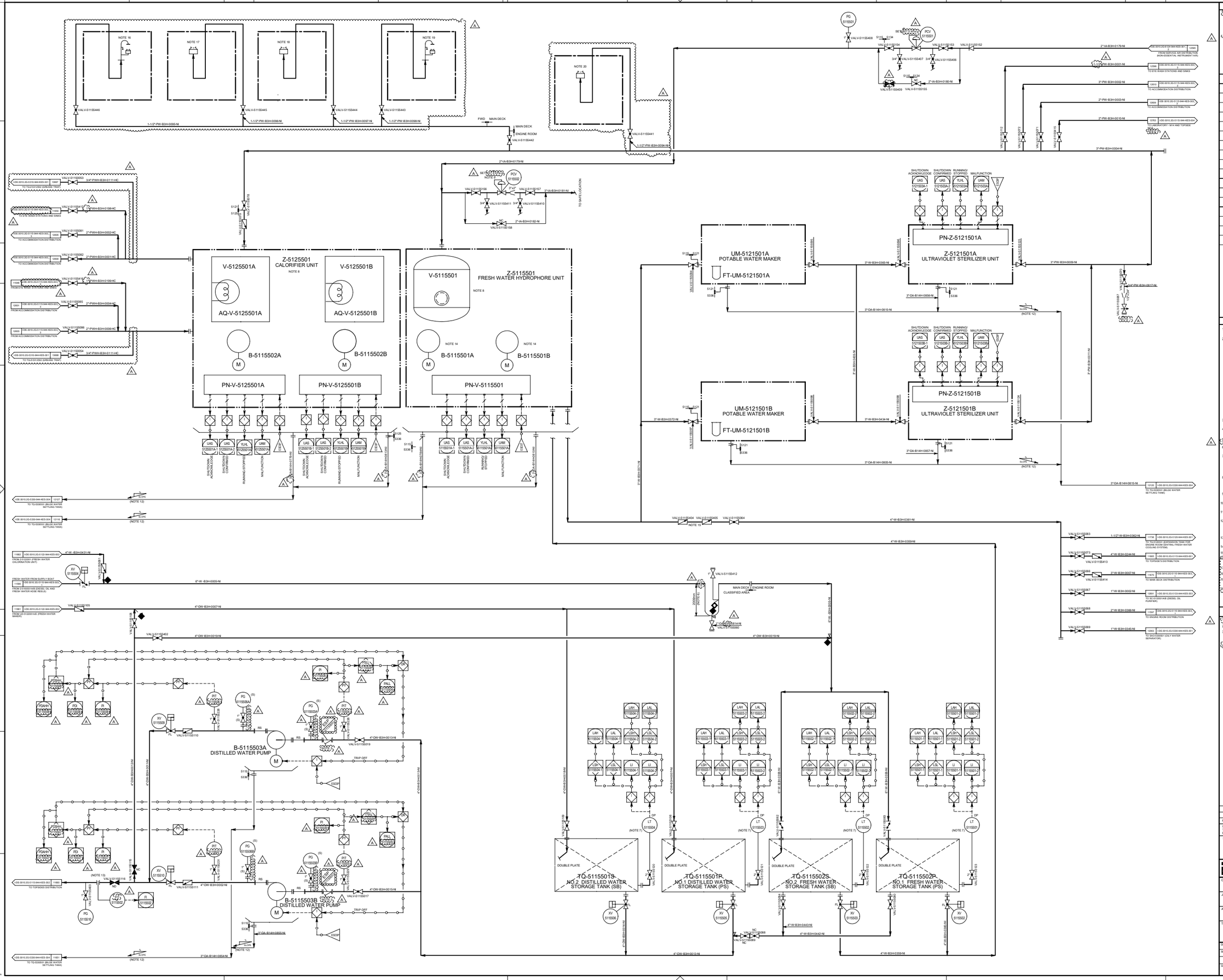

REFERENCE DOCUMENTS

- I-DE-3010.2G-1200-944-KES-001 - SYSTEMS OPERATION PHILOSOPHY
- I-ET-3010.2G-1351-200-KES-301 - PIPING SPECIFICATION FOR HULL
- I-ET-3000.00-0000-940-P4X-002 - SYMBOLS FOR PRODUCTION UNITS DESIGN
- I-ET-3000.00-1200-940-P4X-001 - TAGGING PROCEDURE FOR PRODUCTION UNITS DESIGN
- I-DE-3010.2G-1200-944-KES-302 - CAPACITY PLAN
- I-ET-3010.00-1200-800-P4X-013 - GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS
- I-DE-3010.2G-1200-942-KES-001 - AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS
- I-ET-3010.00-1350-940-P4X-001 - SYSTEMS OPERATION PHILOSOPHY
- I-ET-3010.1M-1200-800-P4X-014 - AUTOMATION INTERFACE OF PACKAGE UNITS
- I-DE-3010.2G-1200-942-KES-318 - GENERAL ARRANGEMENT
- I-ET-3010.1M-1350-200-P4X-001 - HULL PIPING PRACTICE

EQUIPMENT

TAG	DESCRIPTION	TYPE	CAPACITY
AQ-V-5121501A/B (2 x 100%)	CALORIFIER ELECTRIC HEATER	-	
B-5115501A/B (2 x 100%)	FRESH WATER HYDROPHORE PUMP	CENTRIFUGAL	80 m³/h
B-5115502A/B (2 x 100%)	ACCOMMODATION HOT FRESH WATER CIRCULATION PUMP	CENTRIFUGAL	6.3 m³/h
B-5115503A/B (2 x 100%)	DISTILLED WATER PUMP	CENTRIFUGAL	70 m³/h
FT-UM-5121501A/B (2 x 100%)	POTABLE WATER MAKER FILTER	-	3.5 m³/h
PN-V-5115501 (1 x 100%)	FRESH WATER HYDROPHORE VESSEL CONTROL PANEL	CONTROL PANEL	-
PN-V-5125501A/B (2 x 100%)	CALORIFIER WATER VESSEL CONTROL PANEL	CONTROL PANEL	-
PN-Z-5121501A/B (2 x 100%)	ULTRAVIOLET STERILIZER CONTROL PANEL	CONTROL PANEL	-
TQ-5115501P (1 x 100%)	NO.1 DISTILLED WATER STORAGE TANK (PS)	STRUCTURAL	524.0 m³
TQ-5115501S (1 x 100%)	NO.2 DISTILLED WATER STORAGE TANK (SB)	STRUCTURAL	543.5 m³
TQ-5115502P (1 x 100%)	NO.1 FRESH WATER STORAGE TANK (PS)	STRUCTURAL	1620.3 m³
TQ-5115502S (1 x 100%)	NO.2 FRESH WATER STORAGE TANK (SB)	STRUCTURAL	1600.8 m³
UM-5121501A/B (2 x 100%)	POTABLE WATER MAKER	FRESH WATER MAKER UNIT	3.5 m³/h
V-5115501 (1 x 100%)	FRESH WATER HYDROPHORE VESSEL	-	22 m³
V-5125501A/B (2 x 100%)	CALORIFIER WATER VESSEL	-	6.3 m³
Z-5121501A/B (2 x 100%)	ULTRAVIOLET STERILIZER UNIT	STERILIZER UNIT	3.5 m³/h

GENERAL NOTES

1 - THE SAMPLE POINTS LOCATION SHALL BE READILY ACCESSIBLE AND THE PIPE 1-DE-3010.2G-5115-944-KES-304 SHALL BE PROVIDED WITH A SAMPLE VALVE.

2 - SYSTEM DATA:
FRESH WATER SYSTEM
A) WORKING PRESSURE: 1002.2 kPag
B) DESIGN PRESSURE: 1202.0 kPag
C) TEST PRESSURE: 1.5 DP = 1803.0 kPag
D) WORKING TEMPERATURE: 60°C
E) DESIGN TEMPERATURE: 90°C

HOT WATER SYSTEM
A) WORKING PRESSURE: 1000 kPag
B) DESIGN PRESSURE: 1200 kPag
C) TEST PRESSURE: 1.5 DP = 1800.0 kPag
D) WORKING TEMPERATURE: 80°C
E) DESIGN TEMPERATURE: 100°C

DISTILLED WATER SYSTEM
A) WORKING PRESSURE: 849.0 kPag
B) DESIGN PRESSURE: 1175.0 kPag
C) TEST PRESSURE: 1.5 DP = 1763.0 kPag
D) WORKING TEMPERATURE: 30°C
E) DESIGN TEMPERATURE: 60°C

3 - FOR VENT AND SOUNDING 1-DE-3010.2G-5415-944-KES-304 - VENTING AND SOUNDING SYSTEM.

4 - QUANTITY AND LOCATION OF DRAINS SHALL BE DEFINED DURING DETAILING DESIGN DEVELOPMENT. IN ADDITION, ALL LOW AND HIGH POINTS SHALL HAVE DRAIN OR VENT PIPES.

5 - STRAINER TO BE USED ONLY AT COMMISSIONING PHASE.

6 - SIPHON SHALL BE INSTALLED AS CLOSE AS POSSIBLE TO ENGINE ROOM ENTRANCE. THE SIPHON SHALL BE TWO METER HIGH, WITH A PURGE LINE.

7 - THE LEVEL TRANSMITTERS LT-5115501/502/503/504 ARE INCLUDED AT TANK LEVEL MONITORING SYSTEM PACKAGE. THESE INSTRUMENTS SHALL HAVE TWO INDEPENDENT OUTPUTS SIGNALS. ONE SHALL BE CONNECTED TO THE TANK LEVEL MONITORING SYSTEM PANEL AND OTHER SHALL BE CONNECTED TO HULL CONTROL SYSTEM (HCS).

8 - A MECHANICAL INTERLOCK SHALL BE INSTALLED IN THE VALVES THAT ALLOW FOR PSV REMOVAL FOR MAINTENANCE, IN ORDER TO ENSURE THEIR CORRECT ORDER OF OPERATION. FOR DETAILS, SEE I-ET-3010.00-1200-800-P4X-013 - GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS.

9 - THE SET PRESSURE OF PCV-5115502 SHALL BE 5% HIGHER THAN SET PRESSURE OF PCV-5115501.

10 - THIS PAID SHALL BE UPDATED DURING THE DETAIL DESIGN PHASE FOLLOWING ALL EQUIPMENT MANUFACTURERS' RECOMMENDATION.

11 - ALL XV VALVES OF THIS P&ID ARE FULLY REPRESENTED ON PAGE 5 OF I-DE-3010.2G-1200-944-KES-001 - GENERAL NOTES.

12 - SLOPE OF GRAVITY LINES SHALL BE AS FOLLOWS:
TRANSVERSE: 1:50;
FORE TO AFT: 1:100;
AFT TO FORE: 1:25.

13 - IN ORDER TO REDUCE THE FLOW OF DISTILLED WATER TO TOPSIDE CONSUMERS THE OPERATION SHALL FORESEEN A RECIRCULATION TO THE DISTILLED WATER TANK.

14 - EACH PUMP SHALL TRIP IN CASE OF HIGH PRESSURE.

15 - DISSIMILAR CHECK VALVE.

16 - SAFEY SHOWER & EYEWASH FOR FWD PANEL ROOM (F108) LOCATED IN FRONT OF THE DOOR.

17 - EYEWASH FOR FWD PAINT SHOP (F107).

18 - EYEWASH FOR FWD PAINT STORE (F108).

19 - SAFEY SHOWER & EYEWASH FOR FWD INERT GAS GENERATOR ROOM (F112) LOCATED IN FRONT OF THE DOOR.

20 - EYEWASH FOR SEWAGE TREATMENT ROOM (E302).

HOLD TABLE

1 - CANCELLED.

3 - THE QUANTITY AND LOCATION OF POTABLE WATER CONSUMERS FOR M-14, TOPSIDE WCS, EYE-WASH AND SAFETY SHOWER TO BE REVISED DURING DETAILED DESIGN PHASE.

REV.	DESCRIPTION	DATE	EXEC.	CHECK	APPROV.
A	APPROVED FOR CONSTRUCTION	7/27/2021	KSJ.HHI	YCC.HHI	CKW
0	FOR CONSTRUCTION	6/1/2021	KSJ.HHI	YCC.HHI	CKW

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FORM OWNED TO PETROBRAS N-381 REV. L

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PETROBRAS | SRGE

CLIENT: E & P

JOB: BUZIOS FIELD AREA DEVELOPMENT

AREA: PETROBRAS 78 (P-78)

TITLE: PIPING AND INSTRUMENT DIAGRAM
FRESH, HOT AND POTABLE WATER SYSTEM

DESIGN	KES	EXEC.	KSJ.HHI	CHECK	YCC.HHI	APPROV.	CKW
SCALE	NO SCALE	DRAWING	5900.0117818.21.2			SHEET	01 of 01
	(I)		SCP78				
DATE	6/1/2021	No.	I-DE-3010.2G-5115-944-KES-302				

A0 1189mm X 841mm